**Worksheet 5.3**

# A world of reactions

Student: ..... Class: .....

| Balancing a chemical equation | Example: Methane gas will burn in air. This is a combustion reaction. This type of reaction produces $CO_2$ and $H_2O$. |
|---|---|
| **Step 1:** Start with the word equation and name all of the reactants and products. | _____ gas + _____ gas ⟶ carbon dioxide + water |
| **Step 2:** Replace the words in the word equation with formulae and rewrite the equation. | _____ gas = $CH_4$ Oxygen gas = ___ (reactants)<br>_____ = $CO_2$ Water vapour = ___ (products)<br>_____ ⟶ _____ |
| **Step 3:** Count the number of atoms of each element (represented by the formulae of the reactants and products). | **Element** / **Reactants** / **Products**<br>C<br>H<br>O |
| **Step 4:** If the number of atoms of each element is the same on both sides of the equation, the equation is already balanced. If not, numbers need to be placed in front of one or more of the formulae to balance the equation. These numbers are called coefficients and they multiply all of the atoms in the formula. | To balance the hydrogen atoms, put a 2 in front of $H_2O$:<br>$CH_4 + O_2 \longrightarrow CO_2 + 2H_2O$.<br>The oxygen atoms can be balanced by putting a 2 in front of the $O_2$ on the reactant side:<br>$CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$.<br>The equation is now balanced. It can be checked by counting the number of atoms of each element on both sides of the new equation.<br>**Element** / **Reactants** / **Products**<br>C<br>H<br>O |
| **Step 5:** Add physical state symbols. | |

| Balancing a chemical equation | Example: Ethene gas will burn in air. This is a combustion reaction. This type of reaction produces $CO_2$ and $H_2O$. |
|---|---|
| **Step 1:** Start with the word equation and name all of the reactants and products. | _____ gas + _____ gas ⟶ carbon dioxide + water |
| **Step 2:** Replace the words in the word equation with formulae and rewrite the equation. | Ethene gas = $C_2H_4$ Oxygen gas = (reactants)<br>_____ = $CO_2$ Water vapour = ___ (products)<br>_____ ⟶ _____ |
| **Step 3:** Count the number of atoms of each element (represented by the formulae of the reactants and products). | **Element** / **Reactants** / **Products**<br>C<br>H<br>O |
| **Step 4:** If the number of atoms of each element is the same on both sides of the equation, the equation is already balanced. If not, numbers need to be placed in front of one or more of the formulae to balance the equation. These numbers are called coefficients and they multiply all of the atoms in the formula. | _____<br>_____<br>_____<br>_____<br>Check the equation is balanced by counting the number of atoms of each element on both sides of the new equation. |
| **Step 5:** Add physical state symbols. | |

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